```
import pandas as pd

df = pd.read_csv('/content/WA_Fn-UseC_-Telco-Customer-Churn.csv')

df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneServi
0	7590-VHVEG	Female	0	Yes	No	1	N
1	5575-GNVDE	Male	0	No	No	34	Y
2	3668-QPYBK	Male	0	No	No	2	Y
3	7795-CFOCW	Male	0	No	No	45	N
4	9237-HQITU	Female	0	No	No	2	Y

5 rows × 21 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   customerID        7043 non-null   object 
 1   gender            7043 non-null   object 
 2   SeniorCitizen     7043 non-null   int64  
 3   Partner           7043 non-null   object 
 4   Dependents        7043 non-null   object 
 5   tenure            7043 non-null   int64  
 6   PhoneService      7043 non-null   object 
 7   MultipleLines     7043 non-null   object 
 8   InternetService   7043 non-null   object 
 9   OnlineSecurity    7043 non-null   object 
 10  OnlineBackup      7043 non-null   object 
 11  DeviceProtection  7043 non-null   object 
 12  TechSupport       7043 non-null   object 
 13  StreamingTV       7043 non-null   object 
 14  StreamingMovies   7043 non-null   object 
 15  Contract          7043 non-null   object 
 16  PaperlessBilling  7043 non-null   object 
 17  PaymentMethod     7043 non-null   object 
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object 
 20  Churn             7043 non-null   object 
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df.isnull().sum()
```

	0
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	11
Churn	0

dtype: int64

```
df = df.dropna()
```

```
df.shape
```

```
(7032, 21)
```

```
df['Churn'].value_counts()
```

Churn	count
No	5163
Yes	1869

dtype: int64

```
df['Churn'].value_counts(normalize=True) * 100
```

Churn	proportion
No	73.421502
Yes	26.578498

dtype: float64

```
df.groupby('Contract')['Churn'].value_counts(normalize=True).unstack()
```

Contract	No	Yes
Month-to-month	0.572903	0.427097
One year	0.887228	0.112772
Two year	0.971513	0.028487

```
df.groupby('InternetService')['Churn'].value_counts(normalize=True).unstack(
```

InternetService	No	Yes
DSL	0.810017	0.189983
Fiber optic	0.581072	0.418928
No	0.925658	0.074342

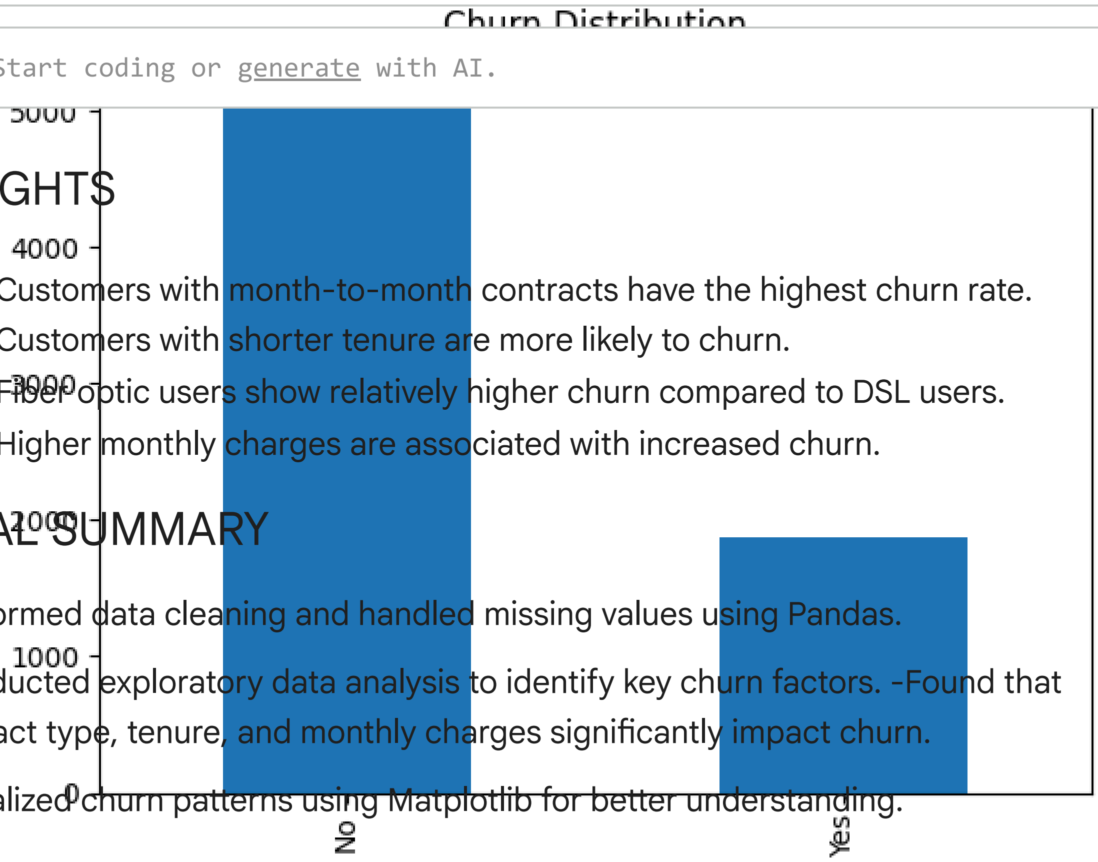
```
df.groupby('tenure')['Churn'].value_counts(normalize=True).unstack()
```

Churn	No	Yes
tenure		
1	0.380098	0.619902
2	0.483193	0.516807
3	0.530000	0.470000
4	0.528409	0.471591
5	0.518797	0.481203
...	...	...
68	0.910000	0.090000
69	0.915789	0.084211
70	0.907563	0.092437
71	0.964706	0.035294
72	0.983425	0.016575

72 rows × 2 columns

```
import matplotlib.pyplot as plt

df['Churn'].value_counts().plot(kind='bar')
plt.title('Churn Distribution')
plt.show()
```



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INSIGHTS

1. Customers with month-to-month contracts have the highest churn rate.
2. Customers with shorter tenure are more likely to churn.
3. Fiber optic users show relatively higher churn compared to DSL users.
4. Higher monthly charges are associated with increased churn.

FINAL SUMMARY

-Performed data cleaning and handled missing values using Pandas.

-Conducted exploratory data analysis to identify key churn factors. -Found that contract type, tenure, and monthly charges significantly impact churn.

-Visualized churn patterns using Matplotlib for better understanding.

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